

## Chapter 2: Discrete-Time Signals and Systems

- ① Signal basics
- ② Discrete-time signals as sequences
- ③ Basic sequences and operations
- ④ Discrete-time systems and LTI systems
- ⑤ Properties of LTI systems (causality, stability, FIR/IIR)
- ⑥ Linear constant-coefficient difference equations (LCCDEs)
- ⑦ Frequency-domain representations and frequency response
- ⑧ Fourier transform of sequences (DTFT)
- ⑨ Symmetry properties and transform theorems
- ⑩ DTFT of special sequences (impulse trains)

## 2.1 Signal Basics

- A signal is a function of one or more independent variables
- 1-D signals are typically functions of time, although other independent variables are equally as valid
- 2-D signals, such as images, are functions of two spatial variables
- The independent variables themselves may be either continuous or discrete
  - A continuous-time signal, often termed an *analog* signal, is defined over a continuum of times
  - A discrete-time signal has independent variable defined only at discrete times

## 2.1 Signal Basics

- A discrete-time signal is further specified to be a *Digital Signal* if the signal values or amplitudes take on discrete values (i.e. finite word-length representations)
- The above signal terminology can be applied to signal processing systems to obtain the following classifications:
  - For a continuous-time system both the input and output are continuous-time signals
  - For a discrete-time system both the input and output are discrete-time signals
  - For a digital system both the input and output are digital signals
- In this course we will deal for the most part with 1-D discrete-time signals

## 2.2 Discrete-time signals as sequences

### Discrete-time signals as sequences

- A discrete-time signal can be represented as a sequence of numbers

$$x_n = x\{n\} = x[n], \quad -\infty < n < \infty.$$

- If  $x[n]$  is obtained by periodic (uniform) sampling of an analog signal  $x_a(t)$ , then

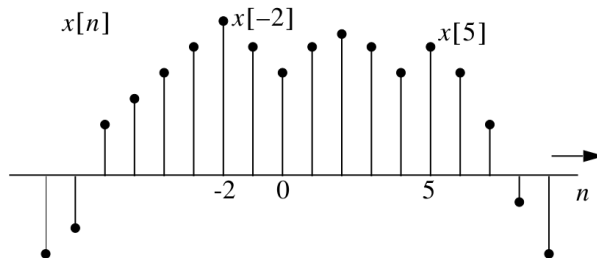
$$x[n] = x_a(nT), \quad -\infty < n < \infty,$$

where  $T$  is the *sample spacing* and  $f_s = 1/T$  is the *sampling frequency (rate)*.

- Later we will show that  $x_a(t)$  can be reconstructed from  $x[n]$  provided the *lowpass sampling theorem* is satisfied.

## 2.2 Discrete-time signals as sequences

**Illustration: sampling an analog waveform**



Discrete-time signal plot.

## 2.3 Basic Sequences and Operations

### Sequence shift (delay/advance)

- Sequence shift or delay is obtained by letting  $n \rightarrow n - n_o$  which defines a new signal

$$y[n] = x[n - n_o]$$

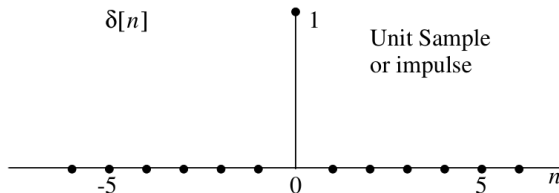
- If  $n_o > 0$  then  $y[n]$  is *delayed* and if  $n_o < 0$  then  $y[n]$  is *advanced*, with respect to  $x[n]$

## 2.3 Basic Sequences and Operations

### Unit sample sequence (impulse)

- Unit sample sequence or impulse,  $\delta[n]$ , is the discrete-time equivalent to  $\delta(t)$ .

$$\delta[n] = \begin{cases} 0, & n \neq 0 \\ 1, & n = 0 \end{cases}$$

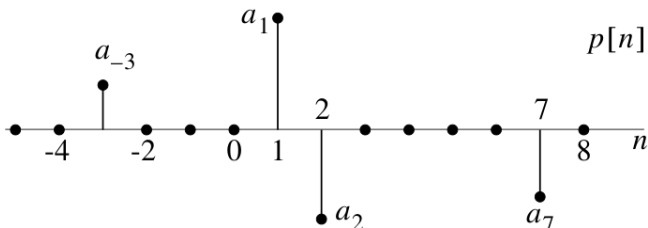




## 2.3 Basic Sequences and Operations

- Any sequence can be written as a linear combination of shifted impulses:

$$x[n] = \sum_{k=-\infty}^{\infty} x[k] \delta[n - k]$$



$$p[n] = a_{-3} \delta[n + 3] + a_1 \delta[n - 1] + a_2 \delta[n - 2] + a_7 \delta[n - 7]$$

## 2.3 Basic Sequences and Operations

### Unit step sequence

- Unit step sequence  $u[n]$

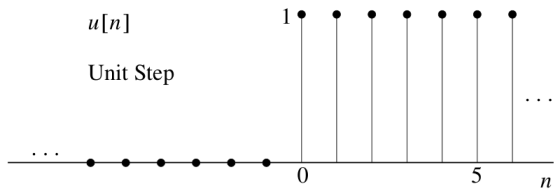
$$u[n] = \begin{cases} 1, & n \geq 0 \\ 0, & n < 0 \end{cases}$$

- It can also be written in terms of shifted impulses:

$$u[n] = \sum_{k=-\infty}^n \delta[k] = \sum_{k=0}^{\infty} \delta[n-k]$$

## 2.3 Basic Sequences and Operations

### Unit step sequence: plot



## 2.3 Basic Sequences and Operations

### Unit step sequence (continuous-time analogy)

- The discrete-time unit step sum is analogous to the continuous-time unit step being written as

$$u(t) = \int_{-\infty}^t \delta(\lambda) d\lambda = \int_0^{\infty} \delta(t - \lambda) d\lambda$$

- Now turn the above relationships around and solve for the impulse function in terms of  $u[n]$ . We first recall that for continuous-time signals

$$\delta(t) = \frac{d}{dt} u(t)$$

- Similarly  $\delta[n]$  can be written as the *first backward difference* of  $u[n]$

$$\delta[n] = u[n] - u[n - 1]$$

## 2.3 Basic Sequences and Operations

### Exponential sequence

$$x[n] = A\alpha^n$$

- where  $A$  and  $\alpha$  may in general be complex constants
  - 1 Suppose  $A$  and  $\alpha$  are both real and  $0 < |\alpha| < 1$ , then  $|x[n]|$  is a decreasing sequence. If  $\alpha < 0$  then the sequence is alternating.
  - 2 Suppose  $A = |A|e^{j\phi}$  and  $\alpha = |\alpha|e^{j\omega_o}$ , then  $x[n]$  is a complex sinusoid with growing, constant, or decaying envelope if  $|\alpha| > 1$ ,  $|\alpha| = 1$ , or  $|\alpha| < 1$ , respectively:

$$x[n] = |A||\alpha|^n e^{j(\omega_o n + \phi)}$$

## 2.3 Basic Sequences and Operations

### Complex sinusoid form

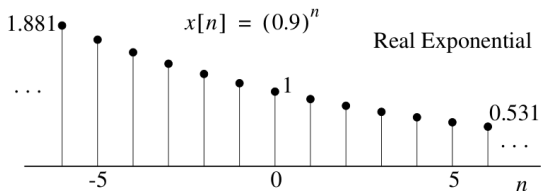
$$x[n] = |A||\alpha|^n [\cos(\omega_o n + \phi) + j \sin(\omega_o n + \phi)]$$

- If  $\alpha = |\alpha|e^{j\omega_o}$ , then  $\alpha^n = |\alpha|^n e^{j\omega_o n}$  combines an exponential envelope and a sinusoid.
- A *real* exponential is the special case  $\omega_o = 0$ .

## 2.3 Basic Sequences and Operations

### Real exponential (example)

$$x[n] = (0.9)^n$$



## 2.3 Basic Sequences and Operations

### Sinusoid sequence

$$x[n] = A \cos(\omega_o n + \phi)$$

- where  $\omega_o$  has units of radians, but may be interpreted as radians/sample if  $n$  has units of samples
- The complex sinusoid, for the case  $\alpha = 1$ , and the real sinusoid both have the property that frequency translates of the form  $\omega_o + 2\pi r$ ,  $r$  an integer, are indistinguishable from one another



## 2.3 Basic Sequences and Operations

### Periodicity

- **Definition:** A sequence  $x[n]$  is periodic with period  $N$  if

$$x[n] = x[n + N], \text{ for all } n$$

- Note that  $N$  must be an integer

## 2.3 Basic Sequences and Operations

### Example 2.2: Periodicity of the sinusoid sequence

- We must establish the conditions under which

$$A \cos(\omega_o n + \phi) = A \cos(\omega_o n + \omega_o N + \phi)$$

- Clearly we must have

$$\omega_o N = 2\pi k \text{ or } \omega_o = \frac{2\pi k}{N}, \text{ } k \text{ an integer}$$

## 2.3 Basic Sequences and Operations

### Example 2.2: Periodicity of the sinusoid sequence (continued)

- Note that the period is not simply  $2\pi/\omega_o$  as in the continuous-time case,  $\omega_o/(2\pi)$  must be rational
- Suppose  $\omega_o = 5\pi/8$ , then the smallest  $N$  that satisfies the above equation is  $N = 16$  ( $k = 5$ )
- Further observe that since  $\omega_o$  and  $\omega_o + 2\pi r$  are indistinguishable frequencies, then a sinusoidal sequence with period  $N$  has the  $N$  distinguishable frequencies

$$\omega_k = \frac{2\pi k}{N}, \quad k = 0, 1, \dots, N - 1$$

on the interval  $[0, 2\pi)$

## 2.3 Basic Sequences and Operations

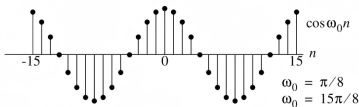
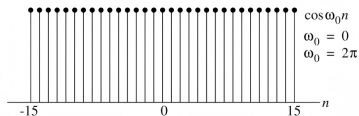
### Example 2.2: Periodicity of the sinusoid sequence (continued)

- For  $x(t) = A \cos(\Omega_o t + \phi)$ , increasing  $\Omega_o$  increases the oscillation rate of  $x(t)$
- For  $x[n] = A \cos(\omega_o n + \phi)$ , increasing  $\omega_o$  increases the oscillation rate of  $x[n]$  only for  $\omega_o \in [0, \pi]$
- For  $\omega_o \in [\pi, 2\pi]$  the oscillation rate decreases back to zero

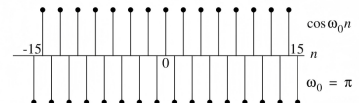
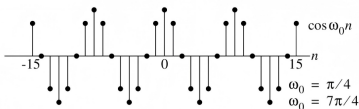
## 2.3 Basic Sequences and Operations

### Plots of $\cos(\omega_0 n)$

Plots of  $\cos(\omega_0 n)$



Plots of  $\cos(\omega_0 n)$



## 2.4 Discrete-Time Systems

### System definition and a motivating example

- Definition: A *discrete-time system* is an operator that maps an input sequence  $x[n]$  into an output sequence  $y[n]$ :

$$y[n] = \mathcal{T}\{x[n]\}.$$

- Example: the *moving-average (MA) operator* (with  $M_1$  samples in the future and  $M_2$  samples in the past)

$$y[n] = \frac{1}{M_1 + M_2 + 1} \sum_{k=-M_1}^{M_2} x[n - k].$$

- Many practical DSP blocks (smoothers, differentiators, predictors) can be written in this form.

## 2.4 Discrete-Time Systems

### Common system properties

- **Memoryless:**  $y[n]$  depends only on  $x[n]$ .
- **Linear:**  $\mathcal{T}\{ax_1[n] + bx_2[n]\} = a\mathcal{T}\{x_1[n]\} + b\mathcal{T}\{x_2[n]\}$ .
- **Time-invariant:** if  $y[n] = \mathcal{T}\{x[n]\}$  then  $\mathcal{T}\{x[n - n_0]\} = y[n - n_0]$ .
- **Causal:**  $y[n_0]$  depends only on  $\{x[n]\}$  for  $n \leq n_0$ .
- **BIBO stable:** bounded input  $\Rightarrow$  bounded output.

Example (nonlinear, memoryless):  $y[n] = \sum_{k=0}^K a_k x^k[n]$ .

## 2.4 Discrete-Time Systems

The class of systems which we are most interested in are those which are both linear and time(shift)-invariant, the so-called **LTI** or **LSI** systems.

- A system is **LTI** if it is **L**inear and **T**ime-**I**nvariant.
- The **unit sample**  $\delta[n]$  is the discrete-time impulse.

- Define the **impulse response**

$$h[n] \triangleq \mathcal{T}\{\delta[n]\}.$$

- Key fact: for an LTI system, knowing  $h[n]$  completely determines the input/output relationship.



## 2.4 Discrete-Time Systems

### Example 2.3: Moving average (impulse response view)

For  $M_1 = 0$  and  $M_2 = M - 1$ ,

$$y[n] = \frac{1}{M} \sum_{k=0}^{M-1} x[n-k].$$

The impulse response is a length- $M$  rectangular pulse:

$$h[n] = \frac{1}{M} (u[n] - u[n-M]).$$

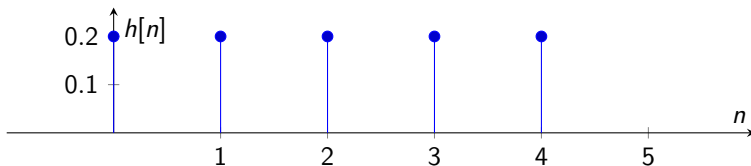


Illustration for  $M = 5$ :  $h[n] = \frac{1}{5}$  for  $n = 0, \dots, 4$  and 0 otherwise.

## 2.4 Discrete-Time Systems

A fundamental result for LTI systems is that the output sequence can be written as the **convolution sum** of the input sequence with the system impulse response.

Recall any sequence can be expanded as a weighted sum of shifted impulses:

$$x[n] = \sum_{k=-\infty}^{\infty} x[k] \delta[n - k].$$

Apply linearity:

$$y[n] = \mathcal{T}\{x[n]\} = \sum_{k=-\infty}^{\infty} x[k] \mathcal{T}\{\delta[n - k]\}.$$

Apply time invariance:  $\mathcal{T}\{\delta[n - k]\} = h[n - k]$  where  $h[n] = \mathcal{T}\{\delta[n]\}$ . Hence

$$y[n] = \sum_{k=-\infty}^{\infty} x[k] h[n - k] \triangleq (x * h)[n].$$

### Example 2.4: Convolve a pulse with an exponential

Let the impulse response be a length- $N$  pulse and the input be a causal exponential:

$$h[n] = u[n] - u[n - N], \quad x[n] = a^n u[n], \quad |a| < 1.$$

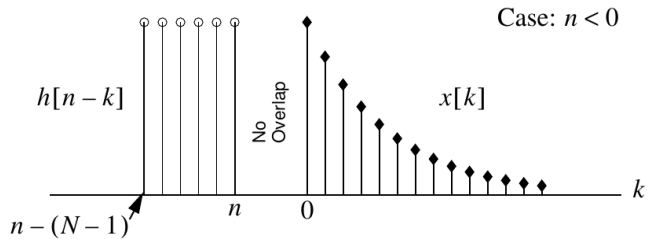
Then

$$y[n] = (x * h)[n] = \sum_{k=-\infty}^{\infty} x[k] h[n - k].$$

Because  $x[k] = a^k u[k]$  and  $h[n - k] = u[n - k] - u[n - k - N]$ , the *limits of the sum* are set by where the two sequences overlap in  $k$ .

## 2.4 Discrete-Time Systems

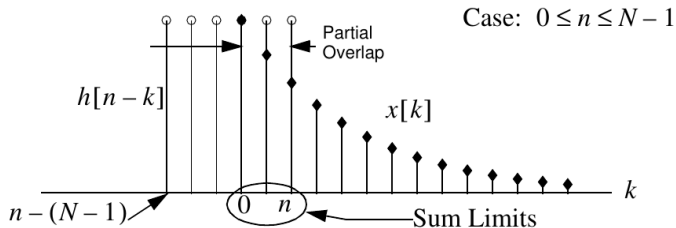
### Example 2.4: Overlap case $n < 0$



No overlap  $\Rightarrow y[n] = 0, \quad n < 0.$

## 2.4 Discrete-Time Systems

**Example 2.4: Overlap case  $0 \leq n \leq N - 1$**

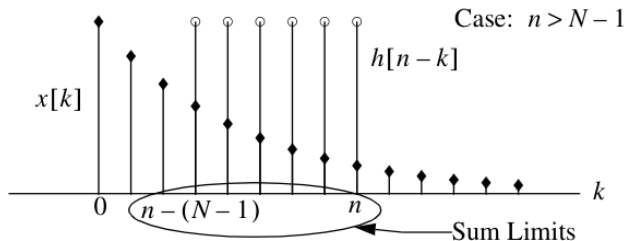


Partial overlap gives the limits  $k = 0, 1, \dots, n$ :

$$y[n] = \sum_{k=0}^n a^k = \frac{1 - a^{n+1}}{1 - a}, \quad 0 \leq n \leq N - 1.$$

## 2.4 Discrete-Time Systems

### Example 2.4: Overlap case $n > N - 1$



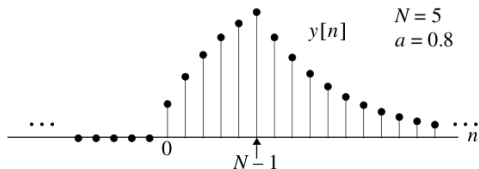
Full overlap (a length- $N$  window) gives the limits  $k = n - (N - 1), \dots, n$ :

$$y[n] = \sum_{k=n-(N-1)}^n a^k = a^{n-N+1} \sum_{k=0}^{N-1} a^k = a^{n-N+1} \left( \frac{1-a^N}{1-a} \right), \quad n > N - 1.$$

## 2.4 Discrete-Time Systems

### Example 2.4: Piecewise result and a plot

$$y[n] = \begin{cases} 0, & n < 0, \\ \frac{1 - a^{n+1}}{1 - a}, & 0 \leq n \leq N - 1, \\ a^{n-N+1} \frac{1 - a^N}{1 - a}, & n > N - 1. \end{cases}$$



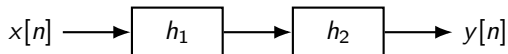
Example plot for  $N = 5$  and  $a = 0.8$ .

## 2.5 Properties of LTI Systems

### Cascade and parallel connections

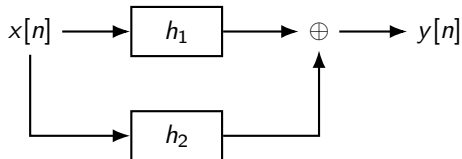
**Cascade:**  $h_1[n]$  followed by  $h_2[n]$

$$h_{\text{eq}}[n] = h_1[n] * h_2[n].$$



**Parallel:**

$$h_{\text{eq}}[n] = h_1[n] + h_2[n].$$



Because convolution is commutative, the cascade order does not matter:  $h_1 * h_2 = h_2 * h_1$ .



## 2.5 Properties of LTI Systems

### BIBO stability theorem (statement)

**Theorem:** An LTI system is BIBO stable *iff*

$$\sum_{k=-\infty}^{\infty} |h[k]| < \infty.$$

Intuition: stability requires the impulse response to be *absolutely summable*.

## 2.5 Properties of LTI Systems

### **BIBO stability proof: sufficiency**

Assume  $\sum_{k=-\infty}^{\infty} |h[k]| = S < \infty$ . For a bounded input  $|x[n]| \leq X_{\max}$ ,

$$|y[n]| = \left| \sum_{k=-\infty}^{\infty} h[k] x[n-k] \right| \leq \sum_{k=-\infty}^{\infty} |h[k]| |x[n-k]| \leq X_{\max} \sum_{k=-\infty}^{\infty} |h[k]| = X_{\max} S.$$

Hence  $y[n]$  is bounded for all  $n$ .

## 2.5 Properties of LTI Systems

### BIBO stability proof: necessity

Assume the system is BIBO stable, but  $\sum_{k=-\infty}^{\infty} |h[k]| = \infty$ .

Define a *bounded* input

$$x[n] = \begin{cases} \frac{h[-n]}{|h[-n]|}, & h[-n] \neq 0, \\ 0, & h[-n] = 0, \end{cases} \quad \Rightarrow \quad |x[n]| \leq 1.$$

Then the output at  $n = 0$  is

$$y[0] = \sum_{k=-\infty}^{\infty} h[k] x[-k] = \sum_{k=-\infty}^{\infty} h[k] \frac{h[k]}{|h[k]|} = \sum_{k=-\infty}^{\infty} |h[k]| = \infty,$$

which contradicts BIBO stability. Therefore  $\sum |h[k]|$  must be finite.

### Causality and FIR/IIR

- **Causal LTI:**  $h[n] = 0$  for  $n < 0$ .
- **FIR (finite impulse response):**  $h[n]$  is nonzero over a finite set of  $n$ .
- **IIR (infinite impulse response):**  $h[n]$  has infinite support (often decays).

Example:  $h[n] = a^n u[n]$  is causal and stable for  $|a| < 1$  but it is IIR.

## 2.5 Properties of LTI Systems

### Example 2.7: impulse, step, and exponential impulse responses

- If  $h[n] = \delta[n]$ , then  $y[n] = x[n]$  (identity system).
- If  $h[n] = u[n]$ , then

$$y[n] = (x * u)[n] = \sum_{k=-\infty}^n x[k]$$

(accumulator / running sum).

- If  $h[n] = a^n u[n]$  with  $|a| < 1$ , then

$$y[n] = \sum_{k=-\infty}^n a^{n-k} x[k]$$

(exponentially weighted smoother).

## 2.6 Linear Constant-Coefficient Difference Equations

A subclass of LTI systems which is of considerable interest are those systems which have input output relationship satisfying an  $N$ th-order constant coefficient difference equation (LCCDE) of the form

$$\sum_{k=0}^N a_k y[n-k] = \sum_{r=0}^M b_r x[n-r].$$

Assuming  $a_0 \neq 0$ , it can be implemented recursively as

$$y[n] = - \sum_{k=1}^N \frac{a_k}{a_0} y[n-k] + \sum_{r=0}^M \frac{b_r}{a_0} x[n-r],$$

together with initial conditions (or “initial rest”).

## 2.6 Linear Constant-Coefficient Difference Equations

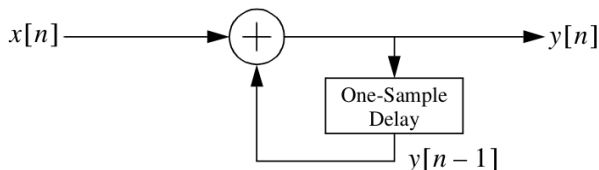
### Example: Accumulator system as an LCCDE

The accumulator is defined by

$$y[n] = \sum_{k=-\infty}^n x[k].$$

It can be equivalently expressed as

$$y[n] - y[n-1] = x[n].$$



For an impulse input  $x[n] = \delta[n]$  and initial rest  $y[n] = 0$  for  $n < 0$ , we have

$$h[n] = u[n].$$

## 2.6 Linear Constant-Coefficient Difference Equations

### Solving LCCDEs: classical method

To solve

$$\sum_{k=0}^N a_k y[n-k] = \sum_{r=0}^M b_r x[n-r],$$

use

$$y[n] = y_h[n] + y_p[n],$$

where:

- $y_h[n]$  solves the homogeneous equation (set  $x[n] = 0$ ),
- $y_p[n]$  is a particular solution for the forcing input.



## 2.6 Linear Constant-Coefficient Difference Equations

### Example 2.10: First-order step response (derivation)

Consider

$$y[n] = a y[n-1] + x[n], \quad n \geq 0,$$

with  $x[n] = K u[n]$ ,  $y[-1] = c$ .

Homogeneous solution:  $y_h[n] = A_1 a^n$ .

Try a particular solution  $y_p[n] = B u[n]$ . Substitute:

$$Bu[n] - aBu[n-1] = Ku[n] \quad \Rightarrow \quad B(1-a) = K \quad \Rightarrow \quad B = \frac{K}{1-a}.$$

Thus

$$y[n] = A_1 a^n + \frac{K}{1-a} u[n].$$

Use  $y[-1] = c$  to solve for  $A_1$  and obtain the full response.

## 2.7 Frequency-Domain Representations

### Eigenfunction property and frequency response

For an LTI system with impulse response  $h[n]$ , complex exponentials are eigenfunctions:

$$x[n] = e^{j\omega n} \Rightarrow y[n] = H(e^{j\omega}) e^{j\omega n},$$

where the **frequency response** is

$$H(e^{j\omega}) = \sum_{n=-\infty}^{\infty} h[n] e^{-j\omega n}.$$

Write

$$H(e^{j\omega}) = H_R(e^{j\omega}) + jH_I(e^{j\omega}) = |H(e^{j\omega})| e^{j\angle H(e^{j\omega})}.$$

By linearity, any linear combination of exponentials maps through the system by scaling each term with  $H(e^{j\omega})$ .

## 2.7 Frequency-Domain Representations

### **Sinusoidal input $\rightarrow$ sinusoidal output**

Let  $x[n] = A \cos(\omega_0 n + \phi) = \Re\{Ae^{j\phi} e^{j\omega_0 n}\}$ . Then

$$y[n] = \Re\{Ae^{j\phi} H(e^{j\omega_0}) e^{j\omega_0 n}\} = A|H(e^{j\omega_0})| \cos(\omega_0 n + \phi + \angle H(e^{j\omega_0})),$$

assuming  $h[n]$  is real (so  $H(e^{-j\omega}) = H^*(e^{j\omega})$ ).

So the system *changes amplitude* by  $|H|$  and *shifts phase* by  $\angle H$ .

## 2.7 Frequency-Domain Representations

### Example 2.13: Ideal delay system

Ideal delay:  $y[n] = x[n - n_d]$ .

Impulse response:  $h[n] = \delta[n - n_d]$ .

Frequency response:

$$H(e^{j\omega}) = \sum_n \delta[n - n_d] e^{-j\omega n} = e^{-j\omega n_d}.$$

Magnitude and phase:

$$|H(e^{j\omega})| = 1, \quad \angle H(e^{j\omega}) = -\omega n_d.$$

## 2.7 Frequency-Domain Representations

### Periodicity of $H(e^{j\omega})$ (proof)

$$H(e^{j(\omega+2\pi)}) = \sum_{n=-\infty}^{\infty} h[n] e^{-j(\omega+2\pi)n} = \sum_{n=-\infty}^{\infty} h[n] e^{-j\omega n} \underbrace{e^{-j2\pi n}}_{=1} = H(e^{j\omega}).$$

Hence  $H(e^{j\omega})$  is always  $2\pi$ -periodic:

$$H(e^{j(\omega+2\pi r)}) = H(e^{j\omega}), \quad r \in \mathbb{Z}.$$

## 2.7 Frequency-Domain Representations

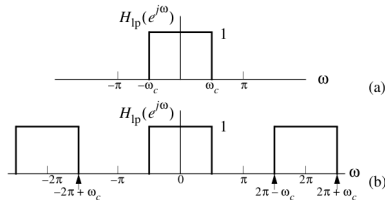
### Example 2.14: Ideal lowpass (continuous-time vs discrete-time)

An ideal analog lowpass has

$$H_{a,lp}(j\Omega) = \begin{cases} 1, & |\Omega| \leq \Omega_c, \\ 0, & |\Omega| > \Omega_c. \end{cases}$$

A discrete-time frequency response must be  $2\pi$ -periodic:

$$H_{lp}(e^{j\omega}) = \begin{cases} 1, & |\omega| \leq \omega_c, \\ 0, & \omega_c < |\omega| \leq \pi, \end{cases} \quad \text{and repeats every } 2\pi.$$



## 2.7 Frequency-Domain Representations

### Example 2.15: Moving average frequency response

For an MA impulse response

$$h[n] = \frac{1}{M_1 + M_2 + 1}, \quad -M_1 \leq n \leq M_2,$$

$$H(e^{j\omega}) = \frac{1}{M_1 + M_2 + 1} \sum_{n=-M_1}^{M_2} e^{-j\omega n} = \frac{1}{M_1 + M_2 + 1} e^{-j\omega \frac{M_2 - M_1}{2}} \frac{\sin\left(\omega \frac{M_1 + M_2 + 1}{2}\right)}{\sin(\omega/2)}.$$

The familiar “ $\sin(Nx)/\sin(x)$ ” envelope is closely related to a sinc shape.

## 2.8 Fourier Transform of Sequences (DTFT)

### DTFT definition and inverse

The discrete-time Fourier transform (DTFT) of  $x[n]$  is

$$X(e^{j\omega}) = \sum_{n=-\infty}^{\infty} x[n]e^{-j\omega n}.$$

The inverse transform is

$$x[n] = \frac{1}{2\pi} \int_{-\pi}^{\pi} X(e^{j\omega}) e^{j\omega n} d\omega.$$

- Any interval of length  $2\pi$  can be used in the inverse integral.
- $X(e^{j\omega})$  is  $2\pi$ -periodic in  $\omega$ .



## 2.8 Fourier Transform of Sequences (DTFT)

### **Proof:** $2\pi$ -periodicity

Starting from the DTFT definition,

$$\begin{aligned}X(e^{j(\omega+2\pi)}) &= \sum_{n=-\infty}^{\infty} x[n]e^{-j(\omega+2\pi)n} \\&= \sum_{n=-\infty}^{\infty} x[n]e^{-j\omega n} \underbrace{e^{-j2\pi n}}_{=1} \\&= \sum_{n=-\infty}^{\infty} x[n]e^{-j\omega n} = X(e^{j\omega}).\end{aligned}$$

Hence  $X(e^{j\omega})$  repeats every  $2\pi$  in  $\omega$ .

## 2.8 Fourier Transform of Sequences (DTFT)

### Existence: absolute summability ( $\ell^1$ condition)

If the sequence is absolutely summable,

$$\sum_{n=-\infty}^{\infty} |x[n]| < \infty \quad (x \in \ell^1),$$

then the DTFT sum converges absolutely for every  $\omega$  and defines a continuous,  $2\pi$ -periodic function.

A useful bound (holds whenever the DTFT converges absolutely):

$$|X(e^{j\omega})| = \left| \sum_n x[n] e^{-j\omega n} \right| \leq \sum_n |x[n]|.$$

So  $\ell^1$  sequences always have a bounded spectrum.

## 2.8 Fourier Transform of Sequences (DTFT)

**Proof sketch:**  $\ell^1 \Rightarrow$  **boundedness and continuity**

**Boundedness:** by the triangle inequality,

$$|X(e^{j\omega})| \leq \sum_n |x[n]| \quad \Rightarrow \quad \sup_{\omega} |X(e^{j\omega})| < \infty.$$

**Continuity:** let  $\omega_k \rightarrow \omega_0$ . For each fixed  $n$ ,  $e^{-j\omega_k n} \rightarrow e^{-j\omega_0 n}$ . Also

$$|x[n]e^{-j\omega_k n}| \leq |x[n]| \quad \text{and} \quad \sum_n |x[n]| < \infty.$$

By dominated convergence, we may pass the limit through the sum:

$$\lim_{\omega_k \rightarrow \omega_0} X(e^{j\omega_k}) = \sum_n x[n] \lim_{\omega_k \rightarrow \omega_0} e^{-j\omega_k n} = \sum_n x[n] e^{-j\omega_0 n} = X(e^{j\omega_0}).$$

So  $X(e^{j\omega})$  is continuous when  $x \in \ell^1$ .

## 2.8 Fourier Transform of Sequences (DTFT)

### **Spectrum: magnitude and phase**

Write the spectrum as

$$X(e^{j\omega}) = X_R(e^{j\omega}) + jX_I(e^{j\omega}) = |X(e^{j\omega})| e^{j\angle X(e^{j\omega})}.$$

- $|X(e^{j\omega})|$  is the magnitude (amplitude) spectrum.
- $\angle X(e^{j\omega})$  is the phase spectrum.
- Phase is defined up to an integer multiple of  $2\pi$  (principal value conventions are common).

## 2.8 Fourier Transform of Sequences (DTFT)

### When $x \notin \ell^1$ : generalized spectra

- If  $x \in \ell^1$ , the DTFT converges absolutely and  $X(e^{j\omega})$  is a continuous,  $2\pi$ -periodic function.
- If  $x \notin \ell^1$  (e.g.,  $u[n]$ , periodic sequences, impulse trains), the DTFT is commonly interpreted using generalized functions (distributions).
- These “non-summable” sequences still have meaningful spectra (line spectra, principal values, impulse terms).

Neote: impulse trains and the unit step are two important examples.

## 2.9 Symmetry Properties and Transform Theorems

### Conjugate symmetry for real-valued sequences

If  $x[n]$  is real-valued, then its DTFT satisfies conjugate symmetry:

$$X(e^{j\omega}) = X^*(e^{-j\omega}).$$

### Consequences:

- $|X(e^{j\omega})|$  is **even** in  $\omega$ .
- $\angle X(e^{j\omega})$  is **odd** in  $\omega$  (up to phase unwrapping).

## 2.9 Symmetry Properties and Transform Theorems

### **Proof: conjugate symmetry**

Start from the DTFT definition:

$$X(e^{j\omega}) = \sum_{n=-\infty}^{\infty} x[n] e^{-j\omega n}.$$

Take the complex conjugate (and use  $x[n] \in \mathbb{R} \Rightarrow x^*[n] = x[n]$ ):

$$X^*(e^{j\omega}) = \left( \sum_{n=-\infty}^{\infty} x[n] e^{-j\omega n} \right)^* = \sum_{n=-\infty}^{\infty} x[n] e^{j\omega n} = \sum_{n=-\infty}^{\infty} x[n] e^{-j(-\omega)n}.$$

Hence

$$X^*(e^{j\omega}) = X(e^{-j\omega}) \quad \Longleftrightarrow \quad X(e^{j\omega}) = X^*(e^{-j\omega}).$$

## 2.9 Symmetry Properties and Transform Theorems

### Even/Odd symmetry (statements)

- If  $x[n]$  is real and even ( $x[n] = x[-n]$ ), then  $X(e^{j\omega})$  is real and even.
- If  $x[n]$  is real and odd ( $x[n] = -x[-n]$ ), then  $X(e^{j\omega})$  is purely imaginary and odd.

Even/odd decomposition (useful often):

$$x_e[n] = \frac{1}{2}(x[n] + x[-n]), \quad x_o[n] = \frac{1}{2}(x[n] - x[-n]).$$



## 2.9 Symmetry Properties and Transform Theorems

### Proof: real-even case

Assume  $x[n] \in \mathbb{R}$  and  $x[n] = x[-n]$ . Then

$$\begin{aligned}X(e^{j\omega}) &= \sum_{n=-\infty}^{\infty} x[n]e^{-j\omega n} \\&= x[0] + \sum_{n=1}^{\infty} x[n]e^{-j\omega n} + \sum_{n=1}^{\infty} x[-n]e^{j\omega n} \\&= x[0] + \sum_{n=1}^{\infty} x[n](e^{-j\omega n} + e^{j\omega n}) \\&= x[0] + 2 \sum_{n=1}^{\infty} x[n] \cos(\omega n).\end{aligned}$$

The result is **real** and depends on  $\cos(\omega n)$ , hence it is **even** in  $\omega$ .

## 2.9 Symmetry Properties and Transform Theorems

### Proof: real-odd case

Assume  $x[n] \in \mathbb{R}$  and  $x[n] = -x[-n]$  (so  $x[0] = 0$ ). Then

$$\begin{aligned} X(e^{j\omega}) &= \sum_{n=-\infty}^{\infty} x[n]e^{-j\omega n} \\ &= \sum_{n=1}^{\infty} x[n]e^{-j\omega n} + \sum_{n=1}^{\infty} x[-n]e^{j\omega n} \\ &= \sum_{n=1}^{\infty} x[n](e^{-j\omega n} - e^{j\omega n}) \\ &= -2j \sum_{n=1}^{\infty} x[n] \sin(\omega n). \end{aligned}$$

The result is **purely imaginary** and depends on  $\sin(\omega n)$ , hence it is **odd** in  $\omega$ .

## 2.9 Symmetry Properties and Transform Theorems

### Time shift (statement + proof)

If  $y[n] = x[n - n_0]$ , then

$$Y(e^{j\omega}) = e^{-j\omega n_0} X(e^{j\omega}).$$

**Proof:**

$$\begin{aligned} Y(e^{j\omega}) &= \sum_{n=-\infty}^{\infty} y[n] e^{-j\omega n} = \sum_{n=-\infty}^{\infty} x[n - n_0] e^{-j\omega n} \\ &= \sum_{m=-\infty}^{\infty} x[m] e^{-j\omega(m+n_0)} = e^{-j\omega n_0} \sum_{m=-\infty}^{\infty} x[m] e^{-j\omega m} = e^{-j\omega n_0} X(e^{j\omega}), \end{aligned}$$

where we substituted  $m = n - n_0$ .

## 2.9 Symmetry Properties and Transform Theorems

### Modulation / frequency shift (statement + proof)

If  $y[n] = e^{j\omega_0 n}x[n]$ , then

$$Y(e^{j\omega}) = X(e^{j(\omega - \omega_0)}).$$

**Proof:**

$$Y(e^{j\omega}) = \sum_{n=-\infty}^{\infty} x[n]e^{j\omega_0 n}e^{-j\omega n} = \sum_{n=-\infty}^{\infty} x[n]e^{-j(\omega - \omega_0)n} = X(e^{j(\omega - \omega_0)}).$$

## 2.9 Symmetry Properties and Transform Theorems

### Convolution theorem (statement)

If

$$y[n] = (x * h)[n] = \sum_{k=-\infty}^{\infty} x[k] h[n - k],$$

then

$$Y(e^{j\omega}) = X(e^{j\omega}) H(e^{j\omega}).$$

## 2.9 Symmetry Properties and Transform Theorems

### Proof: convolution theorem

$$\begin{aligned} Y(e^{j\omega}) &= \sum_{n=-\infty}^{\infty} y[n]e^{-j\omega n} = \sum_{n=-\infty}^{\infty} \left( \sum_{k=-\infty}^{\infty} x[k]h[n-k] \right) e^{-j\omega n} \\ &= \sum_{k=-\infty}^{\infty} x[k] \sum_{n=-\infty}^{\infty} h[n-k]e^{-j\omega n} \quad (\text{swap sums}) \\ &= \sum_{k=-\infty}^{\infty} x[k] \sum_{m=-\infty}^{\infty} h[m]e^{-j\omega(m+k)} \quad (m = n - k) \\ &= \left( \sum_{k=-\infty}^{\infty} x[k]e^{-j\omega k} \right) \left( \sum_{m=-\infty}^{\infty} h[m]e^{-j\omega m} \right) \\ &= X(e^{j\omega}) H(e^{j\omega}). \end{aligned}$$

## 2.9 Symmetry Properties and Transform Theorems

### Parseval / energy theorem (statement)

For finite-energy sequences,

$$\sum_{n=-\infty}^{\infty} |x[n]|^2 = \frac{1}{2\pi} \int_{-\pi}^{\pi} |X(e^{j\omega})|^2 d\omega.$$

This connects time-domain energy to spectral energy density.

## 2.9 Symmetry Properties and Transform Theorems

### Proof: Parseval's relation

Use the inverse DTFT:

$$\begin{aligned}x[n] &= \frac{1}{2\pi} \int_{-\pi}^{\pi} X(e^{j\omega}) e^{j\omega n} d\omega. \\ \sum_{n=-\infty}^{\infty} |x[n]|^2 &= \sum_{n=-\infty}^{\infty} x[n] x^*[n] \\ &= \sum_n \left( \frac{1}{2\pi} \int_{-\pi}^{\pi} X(e^{j\omega}) e^{j\omega n} d\omega \right) \left( \frac{1}{2\pi} \int_{-\pi}^{\pi} X^*(e^{j\theta}) e^{-j\theta n} d\theta \right) \\ &= \frac{1}{(2\pi)^2} \int_{-\pi}^{\pi} \int_{-\pi}^{\pi} X(e^{j\omega}) X^*(e^{j\theta}) \left( \sum_{n=-\infty}^{\infty} e^{j(\omega-\theta)n} \right) d\omega d\theta.\end{aligned}$$

Using the identity

$$\sum_{n=-\infty}^{\infty} e^{j(\omega-\theta)n} = 2\pi \sum_{k=-\infty}^{\infty} \delta(\omega - \theta + 2\pi k),$$

and noting  $\omega, \theta \in [-\pi, \pi]$  leaves only the  $k = 0$  term, so the sum becomes  $2\pi\delta(\omega - \theta)$ . Therefore,

$$\sum_{n=-\infty}^{\infty} |x[n]|^2 = \frac{1}{2\pi} \int_{-\pi}^{\pi} |X(e^{j\omega})|^2 d\omega.$$



## 2.9 Symmetry Properties and Transform Theorems

### Transform theorems (compact summary)

| Time-domain operation         | DTFT-domain effect                 |
|-------------------------------|------------------------------------|
| $y[n] = ax_1[n] + bx_2[n]$    | $Y = aX_1 + bX_2$                  |
| $y[n] = x[n - n_0]$           | $Y = e^{-j\omega n_0} X$           |
| $y[n] = e^{j\omega_0 n} x[n]$ | $Y(\omega) = X(\omega - \omega_0)$ |
| $y[n] = x^*[n]$               | $Y(\omega) = X^*(-\omega)$         |
| $y[n] = x[-n]$                | $Y(\omega) = X(-\omega)$           |
| $y[n] = (x * h)[n]$           | $Y = XH$                           |

(Here  $X(\omega)$  is shorthand for  $X(e^{j\omega})$ .)

## 2.9 Symmetry Properties and Transform Theorems

### Useful DTFT transform pairs (1/2)

| Sequence $x[n]$                  | DTFT $X(e^{j\omega})$                                                                                           |
|----------------------------------|-----------------------------------------------------------------------------------------------------------------|
| 1. $\delta[n]$                   | 1                                                                                                               |
| 2. $\delta[n - n_0]$             | $e^{-j\omega n_0}$                                                                                              |
| 3. 1, $(-\infty < n < \infty)$   | $2\pi \sum_{k=-\infty}^{\infty} \delta(\omega + 2\pi k)$                                                        |
| 4. $a^n u[n]$ , $ a  < 1$        | $\frac{1}{1 - ae^{-j\omega}}$                                                                                   |
| 5. $u[n]$                        | $\text{PV} \left\{ \frac{1}{1 - e^{-j\omega}} \right\} + \pi \sum_{k=-\infty}^{\infty} \delta(\omega + 2\pi k)$ |
| 6. $(n + 1)a^n u[n]$ , $ a  < 1$ | $\frac{1}{(1 - ae^{-j\omega})^2}$                                                                               |

## 2.9 Symmetry Properties and Transform Theorems

### Useful DTFT transform pairs (2/2)

| Sequence $x[n]$                                                              | DTFT $X(e^{j\omega})$                                                                                                                       |
|------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------|
| 7. $\frac{r^n \sin(\omega_p(n+1))}{\sin(\omega_p)} u[n], \quad  r  < 1$      | $\frac{1}{1 - 2r \cos(\omega_p) e^{-j\omega} + r^2 e^{-j2\omega}}$                                                                          |
| 8. $\frac{\sin(\omega_c n)}{\pi n}$                                          | $\begin{cases} 1, &  \omega  < \omega_c, \\ 0, & \omega_c <  \omega  < \pi, \end{cases}$                                                    |
| 9. $\begin{cases} 1, & 0 \leq n \leq M \\ 0, & \text{otherwise} \end{cases}$ | $e^{-j\omega M/2} \frac{\sin(\omega(M+1)/2)}{\sin(\omega/2)}$                                                                               |
| 10. $e^{j\omega_0 n}$                                                        | $2\pi \sum_{k=-\infty}^{\infty} \delta(\omega - \omega_0 + 2\pi k)$                                                                         |
| 11. $\cos(\omega_0 n + \phi)$                                                | $\pi \sum_{k=-\infty}^{\infty} \left[ e^{j\phi} \delta(\omega - \omega_0 + 2\pi k) + e^{-j\phi} \delta(\omega + \omega_0 + 2\pi k) \right]$ |

## 2.9 Symmetry Properties and Transform Theorems

### Example 2.20: DTFT of a delayed exponential

Let

$$x[n] = a^n u[n - 5], \quad |a| < 1.$$

Rewrite and use the time-shift property:

$$x[n] = a^5 a^{n-5} u[n - 5] \implies X(e^{j\omega}) = a^5 e^{-j5\omega} X_0(e^{j\omega}),$$

where  $X_0(e^{j\omega})$  is the DTFT of  $a^n u[n]$ .

$$X(e^{j\omega}) = \frac{a^5 e^{-j5\omega}}{1 - ae^{-j\omega}}, \quad |a| < 1.$$

## 2.10 DTFT of Special Sequences

### 2.10.1 Zero padded impulse train

The sequence of interest is a periodic impulse train (period  $M$ ):

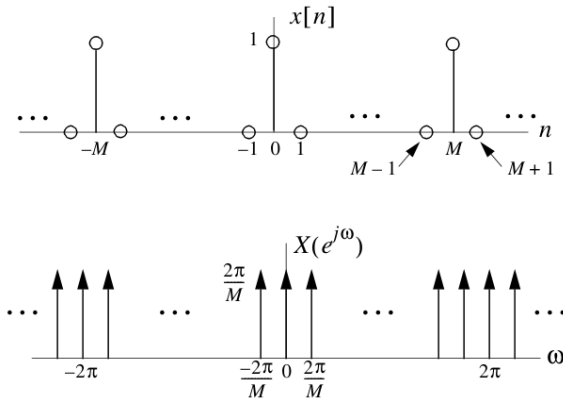
$$x[n] = \sum_{k=-\infty}^{\infty} \delta[n - Mk], \quad M \in \mathbb{Z}^+.$$

- The DTFT is a line spectrum (Dirac impulses).
- Spacing in frequency is  $2\pi/M$ .

$$\sum_{k=-\infty}^{\infty} \delta[n - Mk] \xleftrightarrow{\mathcal{F}} \frac{2\pi}{M} \sum_{k=-\infty}^{\infty} \delta\left(\omega - \frac{2\pi k}{M}\right)$$

## 2.10 DTFT of Special Sequences

### 2.10.1 Zero padded impulse train: illustration



## 2.10 DTFT of Special Sequences

### Derivation: impulse train $\longleftrightarrow$ line spectrum

The impulse train

$$x[n] = \sum_{k=-\infty}^{\infty} \delta[n - Mk]$$

is **periodic** with period  $M$ . Any periodic sequence can be written as a discrete-time Fourier series (DTFS):

$$x[n] = \sum_{m=0}^{M-1} c_m e^{j\frac{2\pi}{M}mn}.$$

The coefficients are

$$c_m = \frac{1}{M} \sum_{n=0}^{M-1} x[n] e^{-j\frac{2\pi}{M}mn}.$$

For the impulse train, only the  $n = 0$  sample in one period is nonzero ( $x[0] = 1$ ), so

$$c_m = \frac{1}{M}, \quad m = 0, 1, \dots, M-1.$$

## 2.10 DTFT of Special Sequences

Therefore

$$x[n] = \frac{1}{M} \sum_{m=0}^{M-1} e^{j\frac{2\pi}{M}mn}.$$

Taking the DTFT and using  $e^{j\omega_0 n} \xleftrightarrow{\mathcal{F}} 2\pi \sum_{k=-\infty}^{\infty} \delta(\omega - \omega_0 - 2\pi k)$ , we obtain

$$X(e^{j\omega}) = \frac{2\pi}{M} \sum_{m=0}^{M-1} \sum_{k=-\infty}^{\infty} \delta\left(\omega - \frac{2\pi m}{M} - 2\pi k\right) = \frac{2\pi}{M} \sum_{k=-\infty}^{\infty} \delta\left(\omega - \frac{2\pi k}{M}\right).$$



- Discrete-time sequences: shifts, impulses, steps, exponentials, sinusoids.
- LTI systems: impulse response and convolution sum.
- Stability/causality: conditions on  $h[n]$ ; FIR vs IIR.
- LCCDEs: time-domain implementations and solution methods.
- Frequency response and DTFT: how systems shape spectra.
- Key transform theorems: shift, modulation, convolution, Parseval.